

MICA — evidence figures

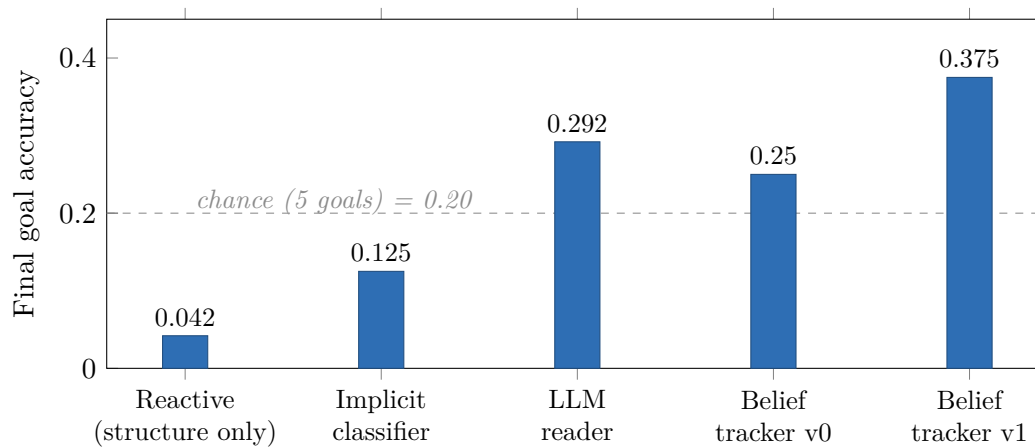


Figure 1: **Intent inference is necessary off-template.** Final goal accuracy on 24 real free-build sessions that no arm ever trained on. The reactive baseline—structure matching alone, the “helper without intent inference”—collapses to 0.042, while the belief tracker (v1) reaches 0.375, a $9\times$ gap. Exactly where builds stop matching templates, structure alone stops working. (Four-arm comparison, exposure-verified splits, cascade of 2026-07-19.)

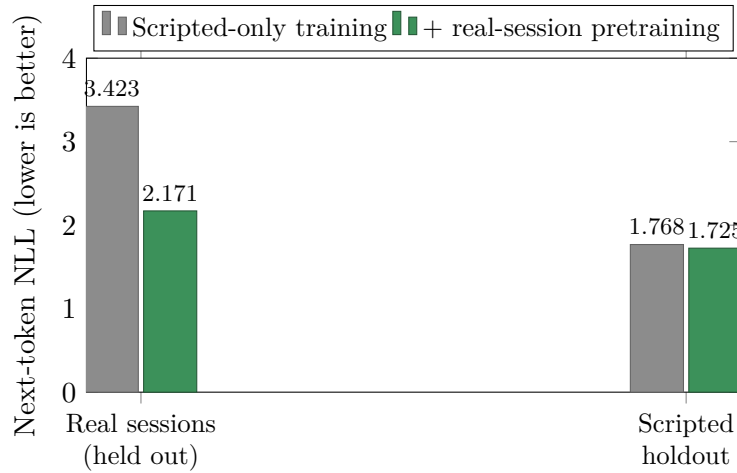


Figure 2: **The decoder’s domain gap, and closing it without labels.** The action decoder trained only on scripted builds is 1.25 nats worse at predicting a real human’s next building action than a scripted one (gray). Pretraining its next-token stage on real sessions’ action streams—usable from *every* session, contested or discarded, because no goal label enters the sample—recovers most of the gap (−37% on held-out real sessions) while slightly improving the scripted holdout. Pre-registered comparison, holdout sessions fixed by id before the run.

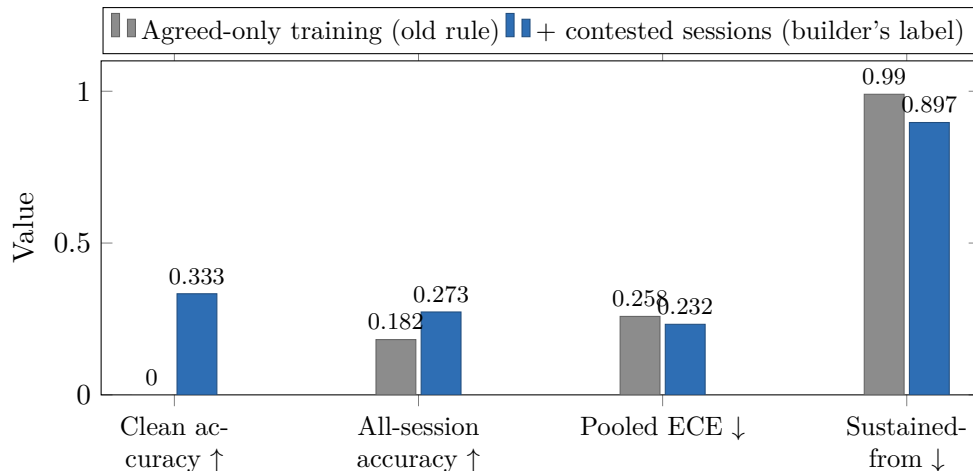


Figure 3: **The builder’s word is good training data.** Training the likelihood heads on matcher-contested sessions—with the builder’s own label as truth—improves every pre-registered criterion on a shared evaluation set neither variant trained on: accuracy on both session groups, calibration (ECE), and *earliness* (sustained-from: the build-progress fraction from which the top goal stays correct; lower = earlier). An identical experiment on 2026-07-12 found the opposite; the reversal followed the repair of a feature leak and the evidence corpus, and is monitored at every retrain. Caveat: the evaluation set is small (11 sessions).

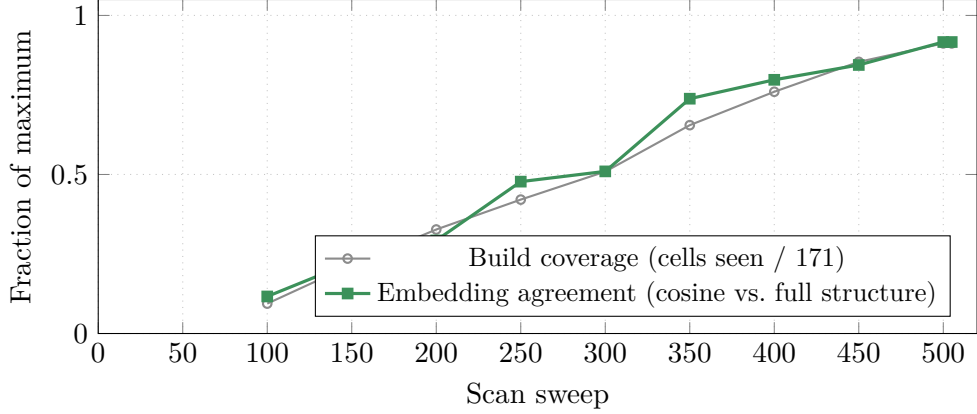


Figure 4: **Structural understanding accumulates with observation.** As the embodied agent’s first-person raycast scan covers more of the human’s build (gray), the 3D shape embedding computed from *only what the agent has seen* converges to the embedding of the complete structure (green, cosine similarity 0.92 at 91% coverage). The agent’s partial view carries the same structural signal as the full ground truth. (Session `fabric-20260719-052914`.)

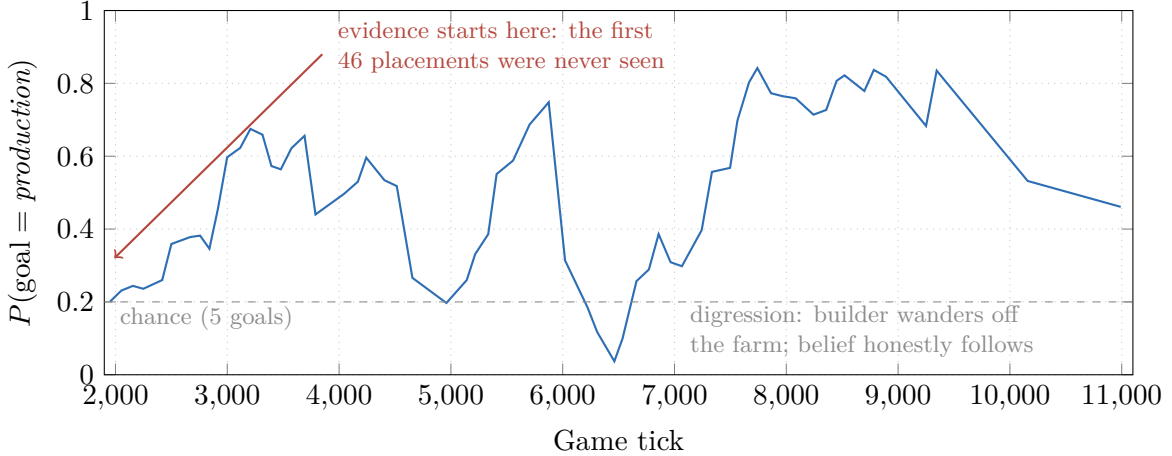


Figure 5: **Case study: tracking a real intention through evidence, absence, and digression.** A live session in which the builder made a crop farm (their own post-session label; goal category *production*). The belief starts at chance because the live pipeline attached late and missed the first 46 placements—yet recovers the correct goal from the remaining evidence, dips when the builder wanders mid-session (an honest response to contradicting evidence, not noise), and re-locks as the farm takes shape, peaking at $P = 0.84$. Illustrative single-session view; quantitative claims rest on the pooled evaluations of Fig. 1.

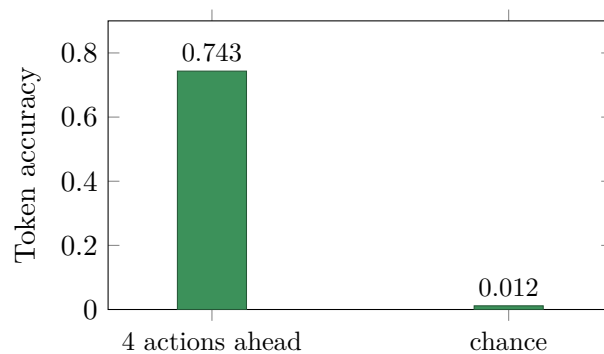


Figure 6: **Anticipation, not just recognition.** The multi-token decoder predicts the builder’s action *four steps ahead* with 74% token accuracy on held-out sessions, against a token chance level of 1.2% (87-token vocabulary, τ -matching: exact for symbolic tokens, ± 2 for coordinates). Understanding an intention here means correctly continuing the build several actions into the future.